



Software Project Management-506

Semester 6th

Name: _____

Roll Number: _____

Computer Science Department
University Of Karachi

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Home Task 3/09/2026: Hands-On Activity: Software Project Management Methods and Tools

Objective: Apply software project management methods and tools to a case study project to understand their practical use in real-world scenarios.

Hardware Requirement

Personal computer, **Laptops or tablets:** For accessing project management tools or templates

Software Requirement

Internet access: For research and collaboration.

Case Study:

Suppose You are a project manager for a software development company. Your team has been assigned to develop an e-commerce platform that will enable users to browse products, add items to their cart, and make payments online. The project has a tight deadline of six months and requires frequent updates due to evolving customer requirements. The company expects rapid development but without compromising on the quality of the software.

Activity 1: Select the Appropriate Project Management Method (15 minutes)

Task:

Based on the case study, choose the most **appropriate project management method** for this software development project. Your options are:

- Waterfall
- Agile
- SCRUM
- Kanban
- Lean
- Critical Path Method (CPM)

Instructions:((Hand Written))

1. **Analyze the project requirements** in the case study.
2. **Identify the key factors** that would influence your choice (e.g., tight deadlines, changing customer needs).



3. **Justify your decision** by explaining **how and why** the chosen method addresses these factors.

Deliverable:

- Submit a hand written one-paragraph explanation for why you chose a particular project management method.
- Highlight the phases or components of the method and how they fit into the project.

Activity 2: Project Planning Using a Tool (30 minutes)

Task:

Using an online project management tool (Trello, Jira, or Microsoft Project), create a project plan for the **e-commerce platform** development.

Instructions:

1. **Choose a tool** from the following options based on your chosen method:
 - **Waterfall:** Use **Microsoft Project** or **Gantt Charts** to plan the project in a sequential manner.
 - **Agile/SCRUM:** Use **Jira** or **Trello** to create backlogs, sprints, and track tasks.
 - **Kanban:** Use **Trello** or **Monday.com** to create task boards with cards for each stage of the workflow.
2. **Create the following components** in your tool: Save and do share screenshots
 - **Project Tasks:** Break down the e-commerce platform development into specific tasks (e.g., “Design User Interface,” “Implement Payment Gateway”).
 - **Timeline:** Assign start and end dates to each task, accounting for dependencies.
 - **Team Roles:** Assign roles and responsibilities to team members (imaginary or real, depending on the class format).
 - **Dependencies:** Identify and link tasks that depend on others (for tools like Gantt Charts or Microsoft Project).
3. **Iterative Planning (Agile/SCRUM):** If using Agile/SCRUM, create at least two sprints (2-3 weeks each) with tasks, and plan for reviews at the end of each sprint.

Deliverable:

- Screenshot or export the project plan from the tool and submit it.
- Provide a brief explanation of how the tool helped structure the project.

Activity 3: Communication and Collaboration Plan (15 minutes)

Task:

Develop a communication plan to manage collaboration among your team members during the project.



Instructions:

1. **Choose a communication tool** (Slack, Microsoft Teams, or Zoom) to manage daily updates, meetings, and discussions.
2. **Define communication protocols** for your team, including:
 - **Daily stand-ups** (if using SCRUM/Agile): Frequency and format.
 - **Weekly meetings**: Schedule for status updates.
 - **Progress tracking**: How will team members update the project manager on task progress? (e.g., via Slack channels or Microsoft Teams).
3. **Outline how you will manage remote collaboration** (if applicable): How will your team members collaborate on code, designs, and documentation if they are working remotely?

Deliverable:

- Submit a 1-page communication plan outlining:
 - The chosen communication tool and why you selected it.
 - Communication protocols and meeting schedules.
 - Collaboration management (especially for remote work).

Activity 4: Time and Risk Management (20 minutes)

Task:

Estimate time requirements for key tasks and identify potential risks in the project.

Instructions:

1. **Use a time tracking tool** like Toggl or Harvest to create time estimates for tasks such as:
 - **Front-end design**
 - **Database setup**
 - **Integration of payment gateway**
 - **Testing and debugging**
2. **Risk Identification**: List at least 3 major risks that could delay the project or impact its quality (e.g., “Payment gateway integration failure” or “Customer requirements changing frequently”).
3. **Risk Mitigation Plan**: For each risk, outline a brief plan on how to address it (e.g., “Have a backup payment gateway in place”).

Deliverable:

- Submit a time estimate for each task, along with a summary of the risks and mitigation strategies.



Activity 5: Reflection and Feedback (10 minutes)

Task:

Reflect on the entire process of applying project management methods and tools to the case study.

Instructions:

1. **What challenges did you face** when applying the selected method to the project?
2. **How did the project management tool** help you organize and track tasks?
3. **What would you do differently** if you were to manage this project in real life?

Deliverable:

- Write a one-paragraph reflection discussing the above points.

Final Submission Guidelines:

- All **activities** must be completed and submitted by **16/09/2024**
- Include screenshots, documents, or any other deliverables as specified in each activity.

Grading Criteria:

- Activity 1: 15 points (Clear choice and justification of method)
- Activity 2: 30 points (Comprehensive project plan using a tool)
- Activity 3: 20 points (Effective communication and collaboration plan)
- Activity 4: 20 points (Accurate time estimates and risk management)
- Activity 5: 15 points (Thoughtful reflection and feedback)